Which Model Changes Can Counterfactual Explanations See? A Ground-Truth Benchmark

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Abstract. Counterfactual methods are used to explain how a classifier changed after an update, but without ground truth one cannot tell which method explains which kind of change. We inject known changes into classifiers on rendered images with exact factors, including a grid that varies the rotation and the offset of the decision boundary, and compare counterfactual differencing, gradient differences, importance differences, disagreement search and differential counterfactuals (Δ-CE), minimal inputs on which exactly one model changes its decision by a margin. A linear analysis predicts that gradient differences see a rotation and are blind to an offset. Both halves were confirmed on CNNs in tests registered before the grid was run (Recovery 0.08 on offsets; +0.57 from the smallest to the largest rotation), and the side of the boundary on which witnesses exist follows the sign of the offset. Margin witnesses abstain on null updates, on rendered images and on a real tabular dataset, where counterfactual differencing attributes a change to about 99% of points. After projection, witnesses and gradient differences coincide as predicted only when the old model's normal is taken at the witness, not at the query (confirmed on new data; seen on two designs). A pre-registered hypothesis in favour of Δ-CE was rejected.

1 Introduction

Deployed classifiers are updated routinely, and an update that improves accuracy can still change individual decisions in unwanted ways [11]. Explaining *how* the decisions of f_{new} differ from those of f_{old} is therefore a natural goal of explainable AI under distributional change [3, 12]. Artelt et al. [1, 2] compare a counterfactual [8] of the same input under each model, or approximate this by a difference of gradients; Muschalik et al. [4] compare feature importances; disagreement search [5, 6, 7] looks for inputs on which two models disagree. These families assume different things about *what* changed (a direction of the weights, a boundary, a region of disagreement), and without ground truth these assumptions go unchecked.

We ask which kind of change each family can see. We inject known changes into classifiers trained on rendered images with exact generating factors and measure which factor each method attributes the change to. We also include *differential counterfactuals* (Δ-CE): the closest input on the data manifold at which only f_{new} changes its decision while f_{old} keeps it, both by a margin κ. Δ-CE anchors disagreement search [5, 6] at a query; unlike robust counterfactuals [9, 10], which seek a point valid for both models, it seeks one valid for exactly

one, and the margin makes it return nothing when two models differ only by jitter.

Contributions. (i) A benchmark with exact ground truth for explaining model updates: five rendered factors, three injected changes, null updates, and a 4×4 grid over the rotation ($\alpha \in \{0, 0.1, 0.25, 0.5\}$) and threshold ($t \in \{0.5, 0.53, 0.56, 0.6\}$) of the decision boundary (49 model pairs), with code and protocol. (ii) A linear analysis that separates rotation from offset and predicts what each family sees. (iii) A test of these predictions on CNNs under a protocol fixed in advance: seven confirmed findings, three failed predictions (one of them, on two designs, explained by a later registered test) and one rejected hypothesis, all reported.

2 Differential counterfactuals and what each family sees

Let $D : \mathcal{Z} \to \mathcal{X}$ map a search space onto the data manifold (here an exact renderer from factor space $\mathcal{Z} = [0,1]^5$). For class y the margin is $m(z) = f_y(D(z)) - f_{1-y}(D(z))$; m_o, m_n belong to f_{old}, f_{new}, and $h = m_n - m_o$. A query z_0 is valid if $m_o(z_0) \geq \kappa$ and $m_n(z_0) \geq \kappa$.

Definition 1. With $\mathcal{F}_\kappa^N = \{z : m_n(z) \leq -\kappa,\ m_o(z) \geq \kappa\}$, a Δ-CE of z_0 is $z^\star \in \arg\min_{z \in \mathcal{F}_\kappa^N} \|z - z_0\|$ if $\|z^\star - z_0\| \leq r$, and NOT_FOUND otherwise; swapping the models gives type O. Each $z \in \mathcal{F}_\kappa^N$ is a *witness*: two forward passes verify that the models decide differently, both decisively, which a difference of two counterfactuals does not.

Lemma 1. Every witness has $h(z) \leq -\kappa - m_o(z) \leq -2\kappa$. Hence models whose margins differ by less than 2κ everywhere have no witness, and if h is L_h-Lipschitz no witness lies within $(h(z_0) + 2\kappa)/L_h$ of z_0. For deep networks L_h is unknown, so this motivates the design rather than certifying it.

Linear case. For linear margins with independent w_o, w_n the witness is unique: the ordinary counterfactual of f_{new} or the minimum-norm point with both constraints active. Write $\hat{w}_n = \cos\theta\, \hat{w}_o + \sin\theta\, v$ with $v \perp \hat{w}_o$, and let d_n, d_o be the distances from z_0 to $\{m_n = -\kappa\}$ and $\{m_o = \kappa\}$. The gradient difference of [1] is $-(\hat{w}_n - \hat{w}_o)$, counterfactual differencing is $u_n - u_o$, the witness is $u^\star$.

Proposition 2. (a) *Offset* ($\theta = 0$): the gradient difference is 0; counterfactual differencing is a multiple of $\hat{w}_o$; a witness exists iff $d_n \leq d_o$ and equals $-d_n \hat{w}_o$. (b) *Rotation* ($0 < \theta < \pi/2$): the gradient difference has components in ratio $\tan(\theta/2) < 1$ along $(\hat{w}_o, v)$, i.e. mainly along the new dependence v; the witness has $|u^\star \cdot \hat{w}_o| = \min\{d_n \cos\theta, d_o\}$ and, for $\theta \leq \pi/4$, is dominated by v only if $d_o(\sin\theta + \cos\theta) < d_n$. After projecting out $\hat{w}_o$ all three coincide, as in Theorem 1 of [1] (on CNNs only with $\hat{w}_o$ at the witness, Section 4.2).

Proofs will be released with the code; on linear models the solver agrees in distance to within 10^{-3} (a stricter pre-declared vector criterion failed at $p = 5$). In words, gradient differences see a *rotation* of the boundary and are blind to a shift of its *offset*; counterfactual differencing and witnesses see the offset, a witness exists only on the side whose margin the offset lowers, and witnesses abstain when the shift is below 2κ, exactly so for a constant logit offset such as

Table 1: Injected changes; $k^\star$ is the ground-truth factor.

Code	Adaptation data	$k^\star$	Pairs
T1 shortcut	colour correlated with label (0.99)	colour	3
T3 subgroup	labels flipped for size > 0.75	size	3
T4 threshold	$y = \mathbb{1}$[aspect > 0.6]	aspect	3
N0 / N1 null	resample / size $\sim U[0.5, 1]$, same rule	—	5 / 3
Grid	$y = \mathbb{1}$[aspect $+ \alpha$(colour $- 0.5) > t$], 4×4	see text	32

the prior correction [13]. The analysis was formulated after the main experiment and before the grid of Section 4.2.

3 Benchmark and protocol

Data and models. Images are 32×32 ellipses rendered from five independent factors in $[0, 1]$: aspect (task factor), size, colour, background and horizontal position; labels are $y = \mathbb{1}$[aspect > 0.5] with 5% noise. The renderer is exact, so the factor change of any explanation is read without a learned oracle. f_{old} is a small CNN (five seeds); f_{new} is f_{old} fine-tuned on 10 000 samples from one condition (Table 1). A pair enters only if its change is visible in decisions on validation data (pre-declared thresholds; null pairs need decisive disagreement ≤ 0.02). For the grid we measure the realised rotation $\hat{\alpha}$ and threshold $\hat{t}$ of each f_{new} by a logistic fit to its decisions, and analyse at these measured values.

Metrics and methods. Per pair we use $N = 300$ valid test points. Recovery@r is the share of points with a valid explanation within the budget whose largest factor change is $k^\star$ (NOT_FOUND and ties count 0); on the grid $k^\star$ is aspect for a pure offset ($\alpha = 0$) and colour otherwise. All counterfactual methods share one solver (Adam with a decaying penalty, then SLSQP on the constrained problem), the same points and the same budget $r = 0.54$, set once on validation. We compare Δ-CE ($\kappa = 0.5$); **B1** counterfactual differencing [1] and a deterministic variant **B1-det**; **B3** gradient difference [1]; **B4** permutation-importance difference [4]; **B5** DIG-style line search [5]; **A1** DeepXplore-style search without proximity [6]; **A2** the ordinary counterfactual of f_{new} alone.

Protocol. Every test split was used once, after all choices were fixed. *Pre-registered:* H2a (Δ-CE beats B3 on T3) and a tie on T1 and T4 ($|\Delta| \leq 0.10$); paired bootstrap stratified by pair. *Confirmatory:* C1–C3, formed after the first results and tested on a new test set (Holm over C1, C3), and H5a–H5d on the grid (Holm, $m = 4$), registered with their thresholds before any grid model was trained; after a pilot without explanation methods the check on $\hat{t}$, the exclusion rule and the statistics were fixed before test data were generated, and one analysis-code error was corrected to the registered rule after the first output; H7c and H8 (Holm, $m = 2$) on new test sets, H8 after a validation run. *Descriptive:* the rest. A tier on coloured digits and a label-shift update failed pre-declared qualification and were stopped.

Table 2: Recovery@r (found-rate) on the locked test split, mean over three pairs; all 95% bootstrap CIs within ±0.035. Last row: found-rate on the 8 null pairs.

	Δ-CE	B1	B1-det	B3	B4	B5	A1	A2
T1	0.38 (0.84)	0.54 (0.93)	0.54 (0.93)	**0.91** (1.0)	0.00	0.02 (0.08)	0.35 (0.63)	0.07 (0.93)
T3	0.53 (0.81)	0.32 (1.0)	0.32 (0.99)	**0.62** (1.0)	0.00	0.06 (0.10)	0.29 (0.58)	0.30 (1.0)
T4	0.47 (0.47)	0.77 (0.90)	0.76 (0.90)	0.07 (1.0)	**1.00**	0.07 (0.08)	0.31 (0.40)	0.90 (0.90)
Null	**0.000**	0.995	0.994	1.0	1.0	0.000	0.000	0.996

4 Results

4.1 The three injected changes

H2a is *rejected*: on T3, Recovery(Δ-CE) − Recovery(B3) = −0.091, 95% CI $[-0.133, -0.050]$, negative in all seeds. Δ-CE beats B1 (+0.21, descriptive) and B5 (+0.47, CI [0.43, 0.50]), which rarely reaches the disagreement region (found-rate ≤ 0.11); we make no solver claim. The predicted tie fails: all four comparisons exceed 0.10 (T1 vs. B3 −0.53, vs. B1-det −0.17; T4 vs. B3 +0.40, vs. B1-det −0.29). B1 and B1-det agree within 0.01 everywhere.

C1 (offset blindness), confirmed: on T4, Recovery(Δ-CE) − Recovery(B3) = +0.31, 95% CI $[0.27, 0.35]$, $p_{Holm} = 2 \cdot 10^{-4}$ (Δ-CE 0.41, B3 0.10, B1-det 0.76). Its content is the failure of B3: on an offset the changed factor is the task factor, so even A2 recovers it (0.90). *C2 (abstention), confirmed:* no witness on any null pair, while B1 attributes a change to 99.1% (N0) and 99.8% (N1) of points; with $\kappa = 0$ witnesses reappear (0.80, 0.89), as Lemma 1 predicts. *C3 (rotation after projection), not confirmed:* on T1, Δ-CE and B3 differ after projecting out $\hat{w}_o$ (0.64 vs. 0.95, 90% CI of the difference $[-0.33, -0.28]$). On T4 every class-1 point has a witness and no class-0 point does (e.g. 136/136 vs. 0/164), as Proposition 2(a) implies.

4.2 The rotation–offset grid

All 32 grid pairs realised the intended rotation; offsets below about 0.05 were not produced reliably ($\hat{t}$ varies by about 0.06 between seeds of a cell). Fig. 1 shows the results.

H5a (offset blindness), confirmed: on pure offsets B3 Recovery is 0.081, 95% CI [0.066, 0.096], below the registered bound 0.20 ($p_{Holm} = 4 \cdot 10^{-4}$); A2 recovers aspect on all of these pairs (0.91–0.98). *H5b (rotation), confirmed:* B3 Recovery rises from 0.36 at $\alpha = 0.1$ to 0.93 at $\alpha = 0.5$, difference +0.57 [0.53, 0.61], $p_{Holm} = 4 \cdot 10^{-4}$; here A2 recovers colour on almost no point (≤ 0.03), so Recovery does separate the methods. *H5c (B3 independent of offset), not confirmed:* adding an offset to a rotation lowers B3 Recovery by about 0.1 (−0.098, 90% CI $[-0.143, -0.053]$, outside the ±0.10 margin). *H5d (C3 on a controlled rotation), not confirmed:* after projection, Δ-CE stays below B3 (−0.35, 90% CI $[-0.38, -0.33]$), as on T1; by the reading fixed in advance, the C3 failure is a failure of the linear analysis on CNNs, not of the shortcut design. Unprojected, Δ-CE recovers the rotation weakly, increasingly with the angle (descriptive).

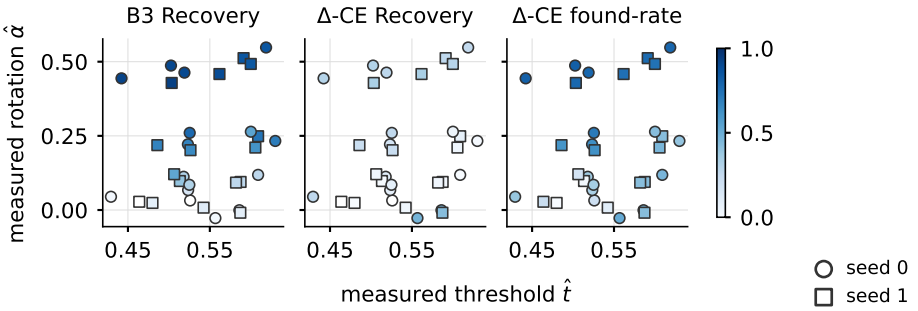


Fig. 1: Grid results at the measured rotation $\hat{\alpha}$ and threshold $\hat{t}$ of each f_{new} (test split): Recovery of B3, Recovery of Δ-CE, and found-rate of Δ-CE, on one colour scale from 0 to 1. The cell $(\alpha, t) = (0, 0.5)$ has no $k^\star$ (hollow markers in the Recovery panels). Marker shape gives the seed of f_{old}.

H8 (normal at the witness), confirmed on new test data for these pairs: with $\hat{w}_o$ taken at the witness instead of z_0, the gap on the Δ-CE support moves from -0.190 to $+0.029$ [0.006, 0.053] ($p_{Holm} = 6 \cdot 10^{-4}$). The normal turns by a median 32° from z_0 to $z^\star$, 27° even for the nearest quarter of witnesses; the registered prediction that this grows with $\|z^\star - z_0\|$ failed. Of the H5d gap, -0.18 is abstention and -0.17 this geometry. On T1: $-0.166 \to -0.024$ (descriptive). C3 and H5d stay not confirmed.

Where witnesses exist (descriptive). On pure offsets witnesses lie only on the side the measured offset lowers, as Proposition 2(a) implies; at shifts of 0.03–0.04 the found-rate depends on the fine-tuning recipe (21–23% of points with the grid recipe, none for a null pair at $+0.035$).

Logit offsets and abstention. For a constant offset b on one logit, witnesses appear exactly at $|b| = 2\kappa$, then on every query of the lowered class only; B1-det attributes a change to 0.961 [0.955, 0.967] of queries already at $|b| \in \{0.5, 0.9\}$, where 3–8% of decisions flip (H7c, $p_{Holm} = 2 \cdot 10^{-4}$). Retraining noise instead fades smoothly with κ. NOT_FOUND is not a solver failure: a search of 10^5 points per query finds no witness on 400 null and 150 impossible-class queries (95% bounds 0.9%, 2.4%), and 3 (4 in a re-run) of 395 on small offsets.

Real tabular data. On five null updates of an MLP on the numerical Adult features (fine-tuning on a new sample, decisive disagreement 0 on validation), Δ-CE returned no witness on 1 500 test points, while B1-det attributed a change to 97.7–100% (registered H6, confirmed).

5 Discussion and limitations

Which family explains a model change depends on the kind of change, and the linear analysis predicts most of it: gradient differences see rotations, increasingly with the angle, and are blind to offsets; witnesses exist on the side the offset moved and abstain on null updates. After projection, witnesses and gradient

differences agree only with the old normal taken at the witness, since on CNNs it turns within the budget; and gradient differences are not fully independent of the offset. A reading not tested as a procedure is to use witnesses to decide *whether* and *where* decisions changed near a query, and gradient differences to describe a rotation.

Limitations. (1) The image data are synthetic; the tabular test covers abstention only (no decoder; integer features treated as continuous). (2) The prior correction [13] is an exact logit offset and behaves as Lemma 1 predicts; a label-shift update by fine-tuning failed qualification and remains untested. (3) κ is an exact threshold only for a constant offset; otherwise abstention tracks decisive decision change, not a calibrated function of a measured shift (Section 4.2). (4) Witnesses see only decision changes on the manifold within the budget, with no certificate for deep networks; our search for missed witnesses is a sampling bound. (5) C1–C3 were formed after the first results; H5a–H5d after the analysis but before the grid. (6) On offsets Recovery rewards any explanation that follows f_{new} (A2), so it measures blindness rather than sensitivity. The protocol had eleven logged amendments and two pre-declared stopping rules; history, proofs and code will be released with the camera-ready version.

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